E-Commerce Customer Churn Analysis

Objective

Analyse customer churn in e-commerce using transactional and behavioural data to uncover key drivers and design actionable retention strategies.

Problem Statement

High churn erodes profitability and loyalty, often driven by poor onboarding, dissatisfaction, payment trust issues, or delivery challenges. Without structured analysis, businesses risk losing valuable customers. This project builds a data-driven framework to identify vulnerable segments and guide proactive retention.

Scope & Deliverables

- **Data Cleaning:** Imputation of missing values, outlier removal.
- **Data Transformation:** Standardization of categorical values, column renaming, derived fields (ComplaintReceived, ChurnStatus).
- **Analysis Queries:** Churn rate by payment mode, tenure, satisfaction, complaints, city tier, and high-risk segments.
- **Dashboard Views:** Modular SQL views (vw_churn_summary, vw_churn_by_payment, etc.) for BI integration.

Key Insights

- **Overall Churn:** Significant proportion of customers churned, requiring urgent retention focus.
- **Payment Mode:** COD customers churn more; UPI/Wallet users are more loyal.
- **Tenure:** Early-stage customers churn quickly; long-tenure churn signals dissatisfaction.
- **Satisfaction:** Low scores strongly predict churn.
- **Complaints:** Complaint-raising customers churn at higher rates.
- **City Tier:** Tier 2/3 cities show higher churn, pointing to logistics challenges.
- **High-Risk Segments:** Male COD users in fashion categories most vulnerable.

Recommendations

- Strengthen **onboarding** for new customers.
- Promote **digital payments** with incentives.
- Enhance **complaint resolution** processes.
- Invest in **regional logistics** for Tier 2/3 cities.
- Launch **segment-specific retention campaigns** for high-risk groups.

Conclusion

The E-Commerce Customer Churn Analysis project showed how structured data cleaning, transformation, and exploratory analysis can reveal key attrition insights. By handling missing values, standardizing categories, and creating derived fields like

ComplaintReceived and ChurnStatus, the dataset became a reliable foundation for churn analysis.